

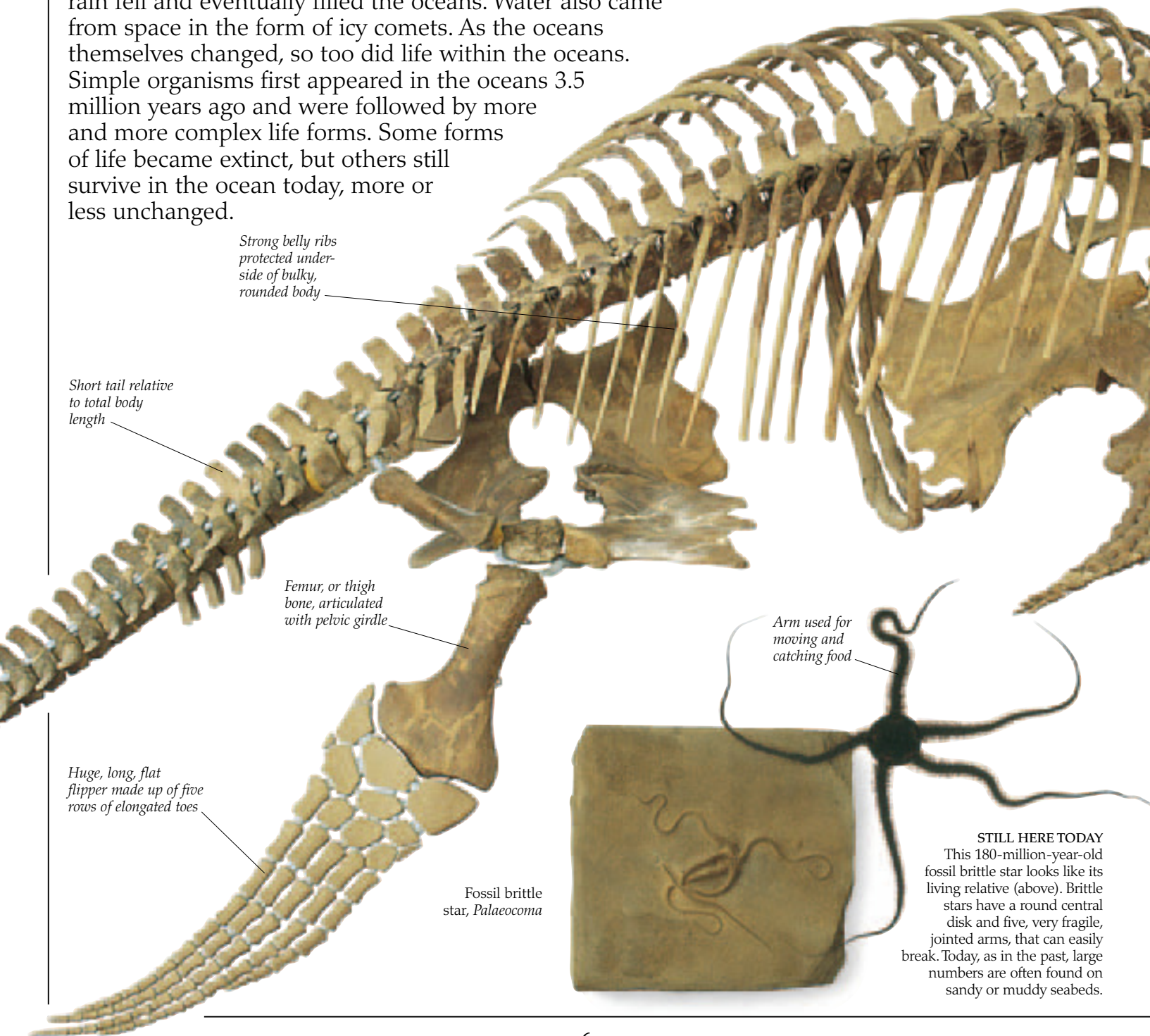
Oceans of the past

THE EARTH, WITH ITS VAST EXPANSES of ocean, has not always looked the way it does today. Over millions of years the land masses have drifted across the face of the planet as new oceans opened up and old oceans have disappeared. Today's oceans only started to take shape in the last 200 million years of the Earth's 4.6-billion-year existence. But water in the form of vapor was present in the atmosphere of the early Earth. As the Earth cooled, water vapor condensed, making storm clouds from which rain fell and eventually filled the oceans. Water also came from space in the form of icy comets. As the oceans themselves changed, so too did life within the oceans. Simple organisms first appeared in the oceans 3.5 million years ago and were followed by more and more complex life forms. Some forms of life became extinct, but others still survive in the ocean today, more or less unchanged.



TOPSY TURVY WORLD

Wiwaxia lived on the seafloor 530 million years ago, yet this fossil was found high above sea level in Canada's Rocky Mountains. This shows just how much the Earth's surface has changed, with land, originally formed under the sea, forced up to form mountain chains.



STILL HERE TODAY

This 180-million-year-old fossil brittle star looks like its living relative (above). Brittle stars have a round central disk and five, very fragile, jointed arms, that can easily break. Today, as in the past, large numbers are often found on sandy or muddy seabeds.